Parallel and Intersecting Lines

CHAPTER
05

Learning Outcomes...

- Plane Surface
- Making Lines with Paper folding
- Intersecting Lines
- Angle
- Perfect Geometry, Imperfect Tools
- Perpendicular and Parallel Lines
- Parallel and Perpendicular Lines in Paper Folding
- Checking Parallel Lines with Transversal
- Transversal of Parallel Lines
- Drawing Parallel Lines by Using Ruler and Set Square
- Making Parallel Lines through Paper Folding

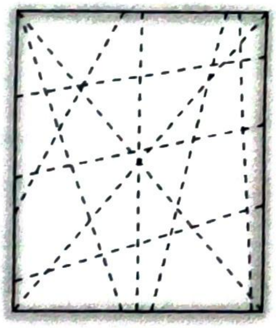
Introduction

Parallel and intersecting lines are two important types of lines that we encounter in our everyday life. Parallel lines never meet, such as the two rails of a railway track or the edges of a notebook. Intersecting lines meet at a point, like roads crossing at a junction or the hands of a clock. In this chapter, you will learn how to recognize, draw, and understand the properties of these lines. This knowledge is useful in designing roads, buildings, artworks, and sports fields. By the end, you will discover how geometry is the part of the world around us.

Plane Surface

A plane surface is a completely flat surface that extends in all directions. It has length and breadth, but no thickness. All the straight lines, shapes and angles we study in geometry are drawn on such a flat surface.

Making Lines with Paper Folding

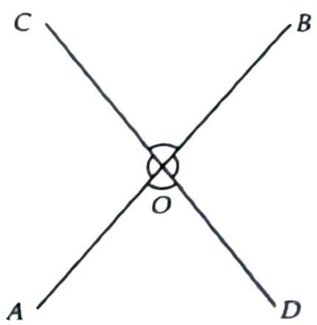


Before we draw lines with a pencil and ruler, let's create them by folding paper. Take a square or rectangular sheet of paper and fold it neatly from one edge to another and then unfold it, we will see a crease is formed where the fold was made. That crease is actually a straight line on the surface of the paper.

If we fold the paper again in another direction, we'll create another line. By keep folding and unfolding the paper in different ways, we'll soon see several lines criss-crossing the surface.

These lines give us a great way to understand different type of lines and relationship between them.

Intersecting Lines



When two folded creases (Two lines) cross each other or meet at a single point, are called intersecting lines.

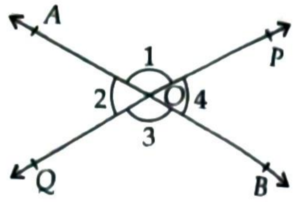
The point where two lines meet or intersect, is called the point of intersection

If you look closely at the adjacent figure, you'll see two lines AB and CD are intersecting each other at O and four angles formed at the point of intersection.

Angle

An angle is the figure formed when two rays start from the same point. This common starting point is called the vertex of the angle.

Vertically Opposite Angles



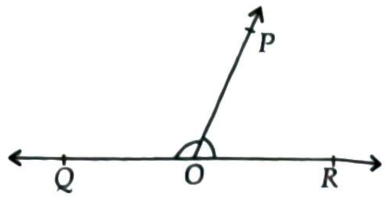
When two lines intersect, they form four angles, each pair of angles formed without any common arm are called vertically opposite angles.

Line AB and PQ intersect at point O forming four angles, $\angle POA$, $\angle AOQ$, $\angle QOB$, $\angle BOP$ and these are labelled as $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ respectively.

Here, $\angle 2$ and $\angle 4$ are without any common arm. So, $\angle 2$ is vertically opposite to $\angle 4$. Similarly, $\angle 3$ is vertically opposite to $\angle 1$.

Note : Vertically opposite angles are equal *i.e.*, $\angle 1 = \angle 3$ and $\angle 2$ and $\angle 4$.

Linear Pair



A linear pair is a pair of adjacent angles, whose non-common arms are opposite rays.

$\angle POQ$ and $\angle POR$ are adjacent angles as O is the common vertex, OP is common arm and their non-common sides are opposite rays. The angles in a linear pair are supplementary.

i.e., $\angle POQ + \angle POR = 180°$.

Perfect Geometry, Imperfect Tools

In geometry, we deal with ideal objects, *i.e.*, lines with no thickness, perfectly measured angles, and accurate shapes. However, when we try to draw these figures in real life using pencils and protractors, small errors often appear. Lines may be too thick, measurements might be slightly off, or the tools we use may not be precise. As a result, angles that should add upto 180°, like linear pairs, may not appear exact. Vertically opposite angles might not seem perfectly equal either. But these differences don't mean the geometric rules are wrong they remind us that geometry is based on logic and reasoning, not just measurement. Even if our drawings aren't perfect, the ideas behind them remain true.

For example ; A straight line forms a 180° angle, so any two angles on it will always add up to 180° and angle around a point is 360°.

COMPETITION WINDOW

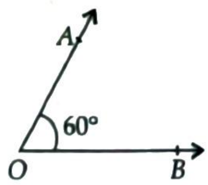
Special Pair of Angles

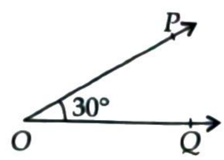
Complementary Angles

- Two angles are said to be complementary, if the sum of their measure is equal to 90°.

  $\angle AOB + \angle POQ = 60° + 30° = 90°$

  $\therefore$ $\angle AOB$ and $\angle POQ$ are complementary angles and $\angle AOB$ is called complement of $\angle POQ$.



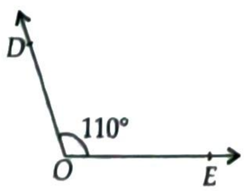


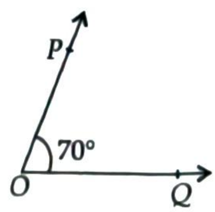
Supplementary Angles

- Two angles are called supplementary, if the sum of their measure is equal to 180°.

  $\angle DOE + \angle POQ = 110° + 70° = 180°$

  $\therefore$ $\angle DOE$ and $\angle POQ$ are supplementary angles and $\angle DOE$ is called supplement of $\angle POQ$.





ILLUSTRATIONS

1 An angle is double its complement. Find the angle.

Soln.: Let the angle be x.

Its complement = double of $x = 2x$

$\therefore \quad x + 2x = 90° \Rightarrow 3x = 90°$

Dividing both sides by 3, we get

$x = \dfrac{90°}{3} = 30°$

$\therefore$ One angle = 30°

and other angle = $2 \times 30° = 60°$

2 Find an angle which is two fifth of its supplement.

Soln.: Let the angle be x.

$\therefore$ Its supplement = $180° - x$

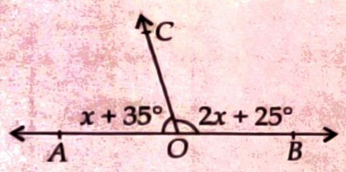
According to question,

$x = \dfrac{2}{5}(180° - x) \Rightarrow 5x = 360° - 2x$

$\Rightarrow 7x = 360° \quad \therefore \quad x = \left(\dfrac{360}{7}\right)^{\circ}$

3 In the given figure, AOB is a straight line. Find

(a) $\angle AOC$ (b) $\angle BOC$



Soln.: Since AOB is a straight line.

$\therefore \quad x + 35° + 2x + 25° = 180°$

$\Rightarrow \quad 3x + 60° = 180°$

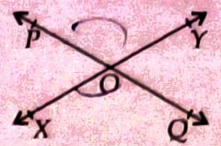
$\Rightarrow \quad 3x = 180° - 60° = 120°$

$\Rightarrow \quad x = \left(\dfrac{120}{3}\right)^{\circ} = 40°$

(a) $\angle AOC = x + 35° = 40° + 35° = 75°$

(b) $\angle BOC = 2x + 25° = 2 \times 40° + 25° = 105°$

4 Identify two pairs of adjacent angles, vertically opposite angles and linear pair in given figure.

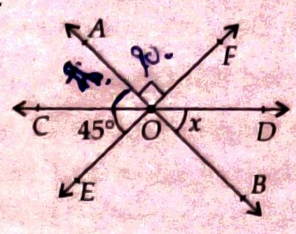


Soln.: Adjacent angles : $\angle POY$, $\angle YOQ$; $\angle YOQ$, $\angle QOX$; $\angle QOX$, $\angle XOP$ and $\angle XOP$, $\angle POY$

Vertically opposite angles : $\angle POY$ and $\angle QOX$; $\angle POX$ and $\angle QOY$.

Linear pair : $\angle QOX$ and $\angle XOP$; $\angle XOP$ and $\angle POY$; $\angle POY$ and $\angle YOQ$; $\angle YOQ$ and $\angle QOX$

5 In the given figure AB, CD and EF are three straight lines intersecting each other at a point O. If $\angle COE = 45°$ and $\angle AOF = 90°$. Find $\angle BOD$.



Soln.: Let $\angle BOD = x$.

$\angle FOD = \angle COE = 45°$

(Vertically opposite angles)

and $\angle AOF = 90°$

Since, AOB is a straight line

$\therefore \quad x + 45° + 90° = 180° \Rightarrow x + 135° = 180°$

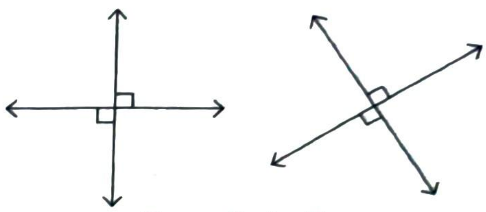
$\Rightarrow \quad x = 180° - 135° = 45° \quad \therefore \quad \angle BOD = 45°$

6 Two roads cross each other to form an "X' at an intersection. One of the angles formed is measured as 112°. What is the angle opposite to it?

Soln.: At the point where two lines intersect, the angles opposite each other are vertically opposite angles and are always equal. So, the opposite angle is also 112°.

Perpendicular Lines

Imagine two lines crossing in such a way that they form perfect L-shapes that means they meet at 90° angles. These are called perpendicular lines. When two lines intersect and all four angles are equal, each angle must be a right angle. For example ; The corners of a square or the edge of your classroom board.

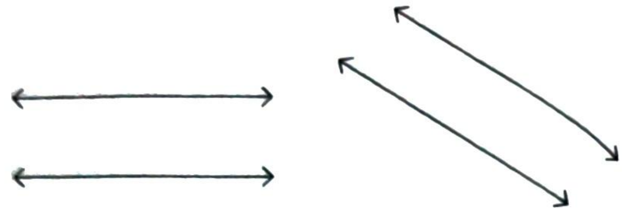


Perpendicular lines

Parallel Lines

Now think of two lines that run side by side and never touch, no matter how long you extend them. These are parallel lines, always the same distance apart, and never intersect.

For example ; Railway tracks, notebook lines or opposite side of a ruler.



Parallel lines

Note : (i) It's important that the perpendicular lines lie on the same flat surface or plane.

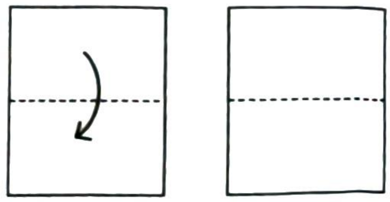
(ii) If a line AB is parallel to line CD and line CD is parallel to line EF, then the line AB is also parallel to the line EF.

Parallel and Perpendicular Lines in Paper Folding

Take a plane square or rectangular sheet of paper without drawing anything on it. We already have a geometric model. The opposite sides of the paper are like railway tracks, they run beside each other and never meet. These are parallel lines. The sides that meet at the corners form perfect right angles, are called perpendicular lines. Now, let's fold the paper to get more lines.

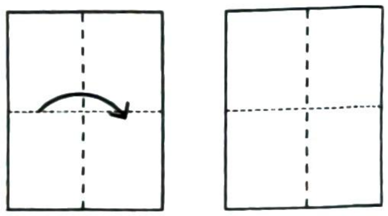
Fold 1 : The horizontal crease

If we fold the paper from top to bottom, aligning the edges carefully and then open it. The crease we get is a new line which is straight and sharp. It runs parallel to the top and bottom edges, and perpendicular to the vertical edges.



Fold 2 : The vertical crease

Now, fold left to right. The new crease crosses the first one at a right angle *i.e.,* exactly 90°. These two lines are perpendicular. Their meeting point forms a corner, just like in a book or window frame.

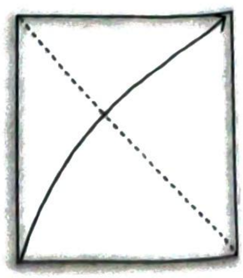


Fold 3 : Patterns of parallel lines

On keep folding in the same direction horizontal or vertical, but in equal steps. Soon, the paper will be filled with parallel lines, all equally spaced. These lines never touch, even if you extend them.

Fold 4 : The diagonal discovery

On folding the paper from one corner to the opposite we get a diagonal line. On folding along the other corners, now we have two diagonals crossing at the center.



These two lines are intersecting lines but these lines have different properties for different shape.

Through folding, we have created intersecting, parallel and perpendicular lines all without drawing a thing.

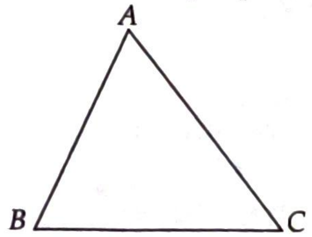
Notations

Mathematicians used special symbols to show relationships between lines :

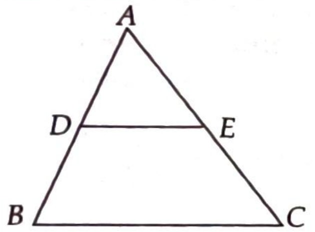
- Parallel lines are marked with an arrow mark (>) or two arrow mark (>>)
- Perpendicular lines are shown with a little square (∟) at the corner :

ILLUSTRATIONS

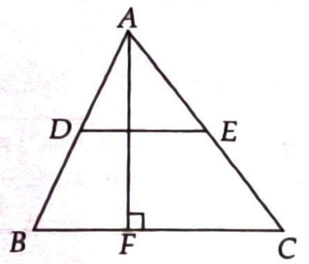
7 In the given triangle draw a line parallel to the base BC and a line perpendicular to BC.



Soln.: Let us draw a line parallel to BC, it will be the staight line drawn by joining midpoint of AB and AC, named as DE.



Now, a line perpendicular to BC can be drawn from point A to BC, meet BC at F.



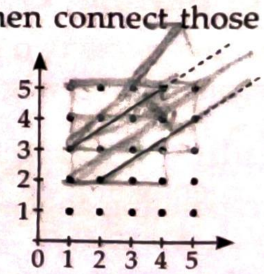
8 Can you draw parallel lines on dot paper that are neither horizontal, vertical nor inclined at 45 degrees? If yes, how can you estimate the distance between them?

Soln.: To draw parallel lines which are neither horizontal, vertical nor inclined at 45 degrees, follow the given steps:

Step-1 : Take a dot grid sheet (graph paper or dot paper with rows and columns of dots)

Step-2 : Pick a starting dot anywhere on the paper. Now; choose a movement pattern. For example; move 3 dots to the right and 2 dots up. Mark a few dots using this pattern, then connect those dots using a ruler to draw your first slanted line.

Step-3 : Now try to draw another line just like the first one, using the same slant. Start from a new dot somewhere else on the paper. Move in the same way, go 3 steps to the right and 2 steps up, then connect those dots.



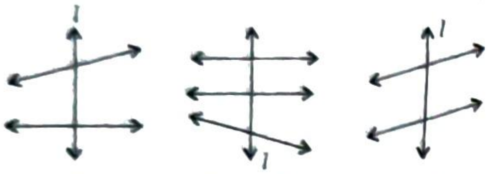
Checking Parallel Lines with Transversal

Sometimes, when we draw slanting lines on a dot grid, especially ones that are not straight across (horizontal), straight down (vertical), or perfectly diagonal (like 45°). It becomes hard to tell if they are truly parallel just by looking. Even if they seem to never meet, we can't be sure just by using our eyes.

To check if two slanting lines are really parallel, we can draw a third line that cuts across both of them. This special line is called a transversal.

Transversals

A line which intersects two or more lines in a plane at distinct points is called a transversal.

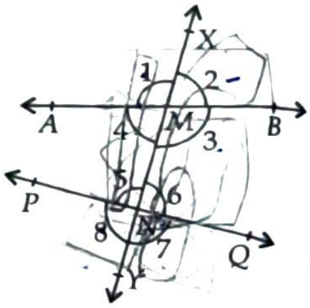
For example : 

Line l is transversal in the three figures as shown above.

Where the transversal crosses the two lines, it creates angles. If the matching angles on both lines are equal, we can say for sure that the two lines are parallel.

When a transversal intersects two or more lines, it forms different kinds of angles with those lines.

Interior Angles : All the angles which have the line segment MN as one of the common arm, are interior angles. $\angle 3$, $\angle 4$, $\angle 5$ and $\angle 6$ are interior angles.



Pair of Alternate Interior Angles : Out of interior angles, the angles that are on opposite sides of transversal and lie in separate linear pairs are pairs of alternate interior angles.

$\angle 4$ and $\angle 6$; $\angle 3$ and $\angle 5$ are pair of alternate interior angles.

Exterior Angles : All angles which do not have the line segment MN as one arm, are exterior angles. $\angle 1$, $\angle 2$, $\angle 7$ and $\angle 8$ are exterior angles.

Pair of Alternate Exterior Angles : Out of the exterior angles, the angles that are on opposite sides of the transversal and lie in separate linear pairs are pairs of alternate exterior angles.

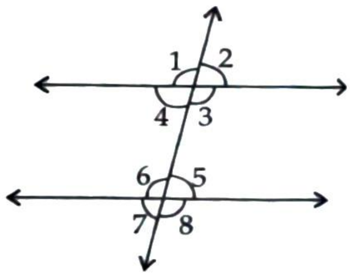
$\angle 1$ and $\angle 7$; $\angle 2$ and $\angle 8$ are pair of alternate exterior angles.

Pair of Corresponding Angles : The pair of angles on one side of the transversal, one of which is an exterior angle while the other is an interior angle but together do not form a linear pair, are pair of corresponding angles. $\angle 1$, $\angle 5$; $\angle 2$, $\angle 6$; $\angle 4$, $\angle 8$; $\angle 3$, $\angle 7$ are pairs of corresponding angles.

Pair of Co-interior Angles : Out of the interior angles, the angles that are on the same side of transversal are pairs of co-interior angles.

$\angle 4$ and $\angle 5$; $\angle 3$ and $\angle 6$ are pairs of co-interior angles.

Transversal of Parallel Lines



When transversal intersects two parallel lines, then

(i) Pair of alternate interior angles are equal *i.e.*,
$\angle 4 = \angle 5$, $\angle 3 = \angle 6$

(ii) Pair of alternate exterior angles are equal *i.e.*, $\angle 1 = \angle 8$, $\angle 2 = \angle 7$

(iii) Pair of corresponding angles are equal *i.e.*,
$\angle 1 = \angle 6$, $\angle 2 = \angle 5$, $\angle 4 = \angle 7$, $\angle 3 = \angle 8$

(iv) Pair of co-interior angles are supplementary *i.e.*,
$\angle 4 + \angle 6 = 180^\circ$, $\angle 3 + \angle 5 = 180^\circ$

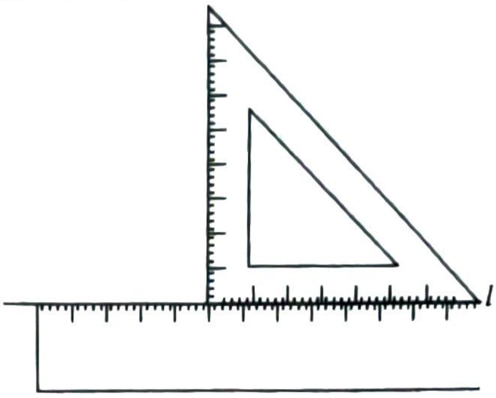
But if the angles don't match, then the lines aren't parallel. So, instead of just guessing, we use the angle relationship formed by the transversal to prove whether lines are parallel or not.

Drawing Parallel Lines by Using Ruler and Set Square

Have you ever used a ruler and set square to draw clean, straight line? Let's discover how these simple tools can help us draw parallel lines and how we can use geometry to prove they are parallel lines.

Step-1 : Use ruler to draw a straight line and name it line l.

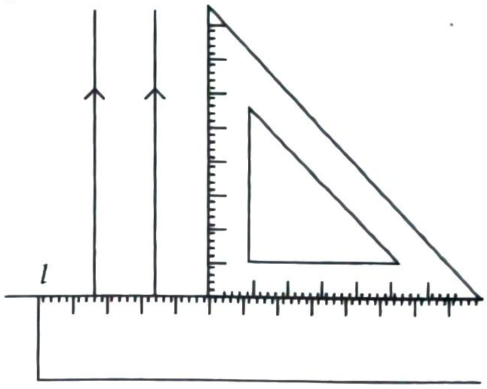
Step-2 : Place one edge of the set square firmly against the ruler so it doesn't move. Make sure the set square forms a right angle (90°) with line l.



Step-3 : Using the other edge of the set square, draw a line perpendicular to line l. This is the first perpendicular line.

Step-4 : Keep the ruler fixed in its position and slide the set square along the ruler without turning it, to another position on the paper.

Step-5 : Again, draw a line along the same edge of the set square. This new line will be parallel to the first one you drew in step 3.



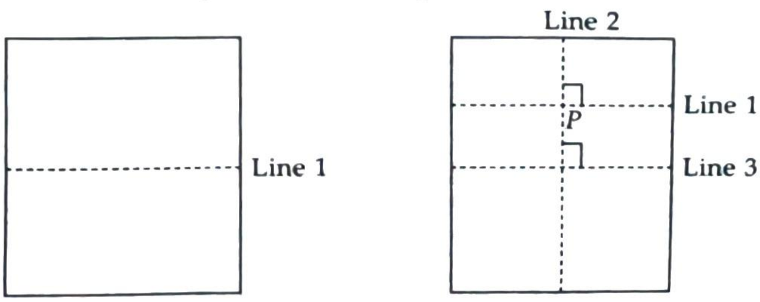
These lines are parallel because they make equal angles with the same line l which acts like a transversal.

Making Parallel Lines through Paper Folding

Take a sheet of paper with a crease call it line 1. Now pick a point P elsewhere on the paper and let's make a new crease through P that's parallel to line 1 by following the two steps:

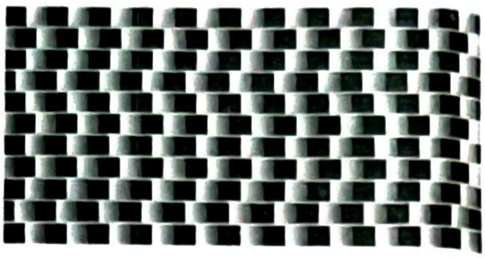
Step-1 : Fold crease through P that's perpendicular to line 1. Call it line 2.

Step-2 : Fold another crease through P that's perpendicular to line 2. Call it line 3.



We get line 1 and line 3 are parallel. As if one line is perpendicular to a second line and that second line is also perpendicular to a third line, then the first and third lines are parallel. It's a neat way to create parallel lines - no ruler or protractor needed.

Parallel Illusions



Although it looks like there are no parallel lines, in reality, many of the lines in these pictures are actually parallel.

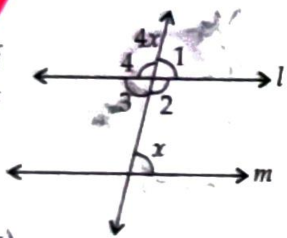
Following are reason why do they look slanted, curved, or tilted.

- Our brain tries to process the whole picture quickly.
- It sometimes makes mistakes, especially when there are strong patterns or contrasting colours.
- This creates a misleading view, even the geometry is perfectly correct.

This illusion is perfect examples of how geometry and visual art blend together. They also remind us why reasoning and proof matter, because our eyes can sometimes be fooled.

ILLUSTRATIONS

9 Find the measure of x in the given figure, if $l \parallel m$.



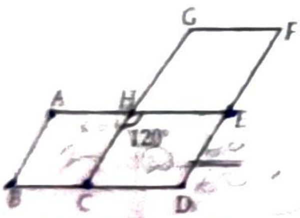
Soln.: As $l \parallel m$ $\therefore$ $\angle 3 = x$
(Alternate interior angles)

Now, $\angle 4 + \angle 3 = 180^\circ$ (Linear pair)

$\angle 4 = 4x$ and $\angle 3 = x$

$4x + x = 180^\circ \Rightarrow 5x = 180^\circ \Rightarrow x = 36^\circ$

10 In the given figure, $AE \parallel GF \parallel BD$, $AB \parallel CG \parallel DF$ and $\angle CHE = 120^\circ$. Find $\angle ABC$ and $\angle CDE$.



Soln.: Since, $AE \parallel BD$ and CH is a transversal.

$\therefore$ $\angle CHE = \angle HCB = 120^\circ$ ...(i)
[Alternate interior angles]

Now, $CH \parallel DF$ and CD is a transversal.

$\therefore$ $\angle HCB = \angle CDE$ [Corresponding angles]

$\Rightarrow$ $\angle CDE = 120^\circ$ ...(ii) [Using (i)]

Also, $AB \parallel DF$ and BD is a transversal.

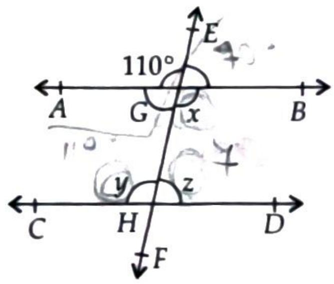
$\therefore$ $\angle ABC + \angle CDE = 180^\circ$ [Co-interior angles]

$\Rightarrow$ $\angle ABC = 180^\circ - 120^\circ$ [Using (ii)]

$= 60^\circ$

Thus, $\angle ABC = 60^\circ$ and $\angle CDE = 120^\circ$

11 In the given figure, $AB \parallel CD$ and EF is a transversal. If $\angle AGE = 110^\circ$, then find x, y and z.



Soln.: We have,

$x = \angle AGE = 110^\circ$ (Vertically opposite angles)

$y = x = 110^\circ$ (Alternate interior angles)

Since, the sum of co-interior angles is 180°.

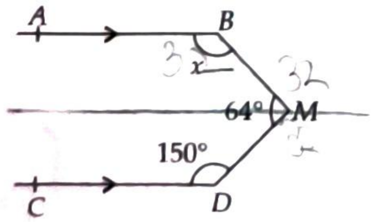
$\therefore$ $x + z = 180^\circ$

$\Rightarrow$ $110^\circ + z = 180^\circ$

$\Rightarrow$ $z = 180^\circ - 110^\circ = 70^\circ$

$\therefore$ $x = 110^\circ$, $y = 110^\circ$ and $z = 70^\circ$

12 In the given figure $AB \parallel CD$, $\angle CDM = 150^\circ$, $\angle BMD = 64^\circ$ and $\angle ABM = x$. Find the value of x.



Soln.: Through M, draw $EMF \parallel CD \parallel AB$

Let $\angle EMB = y$

Then, $\angle EMD = 64° - y$

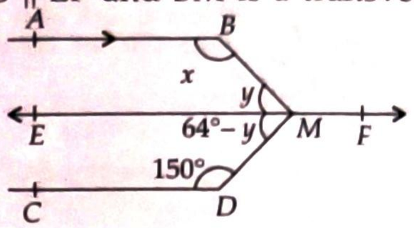
Now, $EF \parallel CD$ and MD is a transversal

$\therefore$ The sum of co-interior angles is 180°.

$\therefore$ $150° + 64° - y = 180°$

$\Rightarrow$ $y = 150° + 64° - 180° = 34°$

Again $AB \parallel EF$ and BM is a transversal.

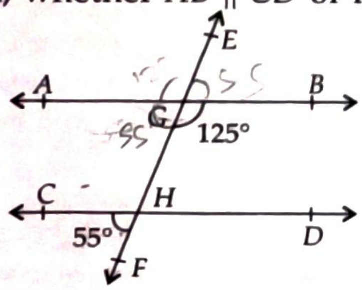


$\therefore$ Sum of co-interior angles is 180°.

$\therefore$ $x + y = 180°$

$\Rightarrow$ $x + 34° = 180°$ $\Rightarrow$ $x = 180° - 34° = 146°$

13 Check, whether $AB \parallel CD$ or not?



Soln.: $\angle GHD = \angle CHF = 55°$

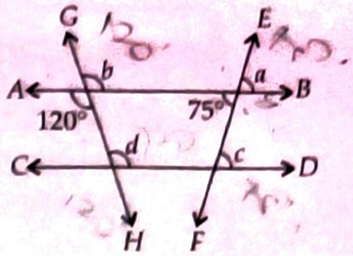
(Vertically opposite angles)

Now, $\angle BGH + \angle GHD = 125° + 55° = 180°$

Thus, the sum of co-interior angles is 180°

Hence, $AB \parallel CD$

14 In the given figure $AB \parallel CD$ and EF and GH are two transversals, find the values of a, b, c, d.



Soln.: $AB \parallel CD$ and EF is a transversal.

$c = 75°$ (Alternate interior angles)

Again, $c = a$ (Corresponding angles)

$\therefore$ $a = 75°$

Now, $AB \parallel CD$ and GH is a transversal.

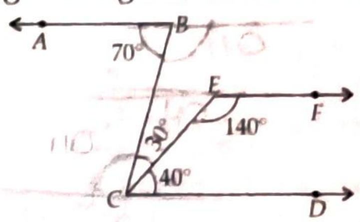
$\therefore$ $d = 120°$ (Alternate interior angles)

Again, $b = d$ (Corresponding angles)

$\Rightarrow$ $b = 120°$

$\therefore$ $a = 75°$, $b = 120°$, $c = 75°$ and $d = 120°$

15 In the given figure, show that $AB \parallel EF$.



Soln.: $\angle ABC = 70°$ (Given) ... (i)

Now, $\angle BCD = \angle BCE + \angle ECD \Rightarrow \angle BCD = 30° + 40°$

$\Rightarrow$ $\angle BCD = 70°$ ...(ii)

From (i) and (ii), we get

$\angle ABC = \angle BCD$

Since, alternate interior angles are equal, therefore, $AB \parallel CD$.

Now, $\angle FEC + \angle ECD = 140° + 40° = 180°$

Since, sum of the interior angles made by the transversal on its same side is 180°.

$\therefore$ $EF \parallel CD$

Since, $AB \parallel CD$ and $EF \parallel CD$

$\therefore$ $AB \parallel EF$

CONCEPT MAP

PARALLEL AND INTERSECTING LINES

Basic concept

Intersecting Lines

Lines that meet or cross at a single point. The point where they meet is called the point of intersection.

Perpendicular Lines

Lines that intersect each other at 90°.
Perpendicular Lines are denoted with a small square symbol at the point of intersection.

Parallel Lines

Two lines on the same plane that never meet, no matter how far extended are called parallel lines.
Parallel Lines are denoted with arrow marks (e.g., >, >>)

Types of Angles

Adjacent Angles

Angles next to each other that share a common side are called adjacent angles.

Vertically Opposite Angles

Angles opposite to each other at the point of intersection of two lines are always equal.

Complementary Angles

Two angles that add up to 90°.

Supplementary Angles

Two angles that add up to 180°.

Linear Pair

Adjacent angles formed by two intersecting lines that add up to 180°.

Transversals

Types of Angles Formed by a Transversal

- Interior Angles
- Exterior Angles
- Corresponding Angles
- Alternate Interior Angles
- Alternate Exterior Angles
- Interior Angles on the same side of the Transversal

Transversal

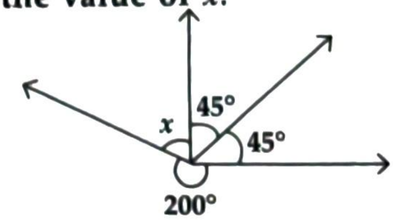
A line that intersects two or more lines at distinct points.

Properties of Angles

- Corresponding angles are equal if the lines are parallel.
- Alternate interior angles are equal if the lines are parallel.
- Interior angles on the same side of the transversal are supplementary if the lines are parallel.

Solved Examples

1. **Find the value of x.**

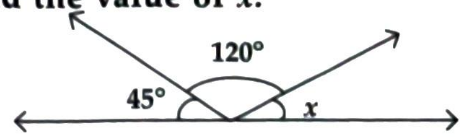


Soln.: $x + 45° + 45° + 200° = 360°$ (Complete angle)

$\Rightarrow \quad x + 290° = 360° \quad \Rightarrow \quad x = 360° - 290°$

$\Rightarrow \quad x = 70°$

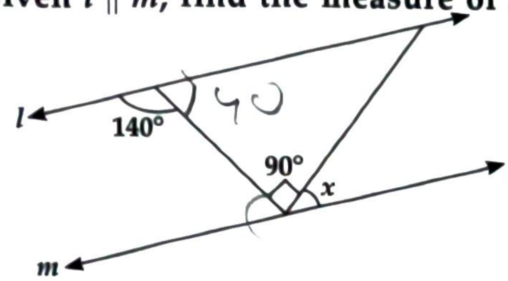
2. **Find the value of x.**



Soln.: $x + 120° + 45° = 180°$ (Linear Pair)

$\Rightarrow \quad x = 180° - 165° \quad \Rightarrow \quad x = 15°$

3. **Given $l \parallel m$, find the measure of x.**

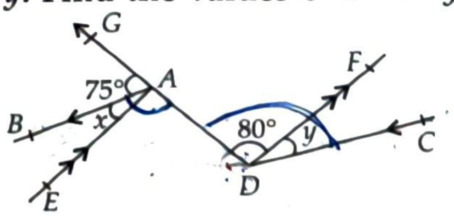


Soln.: Since $l \parallel m$

$\therefore \quad 140° = 90° + x$ (Alternate interior angles)

$\Rightarrow \quad 140° - 90° = x \Rightarrow x = 50°$

4. **In the given figure, $AB \parallel CD$ and $EA \parallel DF$. If $\angle GAB = 75°$, $\angle ADF = 80°$, $\angle EAB = x$ and $\angle CDF = y$. Find the values of x and y.**



Soln.: $EA \parallel DF$ and AD is transversal.

$\therefore \quad \angle EAD = \angle ADF = 80°$ (Alternate interior angles)

Now, $\angle GAB + \angle EAB + \angle EAD = 180°$ (Linear pair)

$\Rightarrow \quad 75° + x + 80° = 180°$

$\Rightarrow \quad x + 155° = 180° \Rightarrow \quad x = 180° - 155° = 25°$

$\therefore \quad \angle BAD = x + 80° = 25° + 80° = 105°$

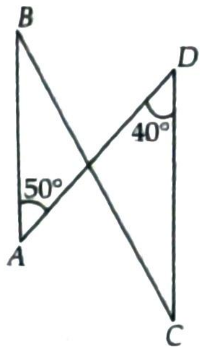
Again $AB \parallel CD$ and AD is a transversal.

$\therefore \quad \angle ADC = \angle BAD$ (Alternate interior angles)

$\Rightarrow \quad 80° + y = 105° \quad \Rightarrow \quad y = 105° - 80° = 25°$

$\therefore \quad x = 25°$ and $y = 25°$

5. **Is $AB \parallel CD$?**



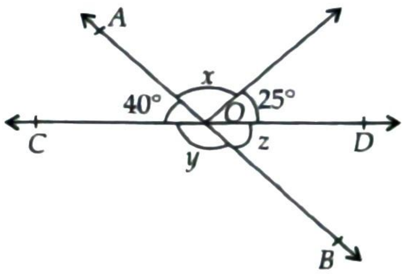
Soln.: Transversal AD cuts two lines AB and CD. So, $\angle BAD$ and $\angle CDA$ is a pair of alternate angles.

But $\angle BAD = 50°$ and $\angle CDA = 40°$

Thus, alternate angles are not equal.

$\therefore \quad AB$ and CD are not parallel to each other.

6. **Find the values of x, y and z.**



Soln.: Since z and $\angle AOC$ are vertically opposite angles.

$\therefore \quad z = 40°$

Now, $40° + x + 25° = 180°$ (Linear pair)

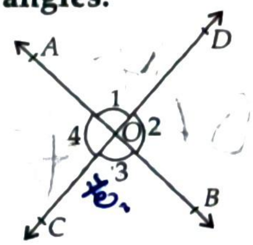
$\Rightarrow \quad x + 65° = 180° \Rightarrow x = 180° - 65° = 115°$

Also, $\quad y + z = 180°$ (Linear pair)

$\Rightarrow \quad y + 40° = 180° \Rightarrow y = 180° - 40° = 140°$

$\therefore \quad x = 115°, y = 140°, z = 40°$

7. **Line AB and CD intersect at O. If $\angle 3 = 70°$. Find all other angles.**



Soln.: Since, CD is a straight line.

$\therefore$ $\angle 2$ and $\angle 3$ form a linear pair of angles.

$\Rightarrow$ $\angle 2 + \angle 3 = 180°$

$\Rightarrow$ $\angle 2 = 180° - 70° = 110°$

Since, $\angle 1 = \angle 3$ (Vertically opposite angles)

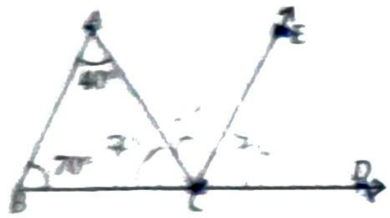
$\therefore$ $\angle 1 = 70°$

Also, $\angle 4 = \angle 2 = 110°$ (Vertically opposite angles)

$\angle 1 = 70°$, $\angle 2 = 110°$, $\angle 3 = 70°$ and $\angle 4 = 110°$

8. In the given figure, side BC of ΔABC has been produced to D. Line $CE \parallel BA$. If $\angle ABC = 70°$, $\angle BAC = 40°$, find

(i) $\angle ACE$ (ii) $\angle ECD$ (iii) $\angle ACD$



Soln.: (i) $CE \parallel BA$ and AC is a transversal.

$\therefore$ $\angle BAC = \angle ACE$ (Alternate interior angles)

Hence, $\angle ACE = 40°$

(ii) $AB \parallel EC$ and BD is a transversal, intersecting the parallel lines at B and C.

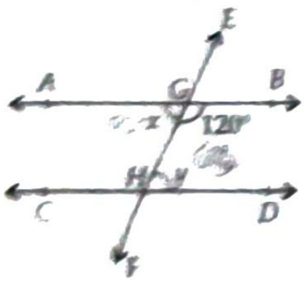
$\angle ABC = \angle ECD$ (Corresponding angles)

$\Rightarrow$ $\angle ECD = 70°$

(iii) $\angle ACD = \angle ACE + \angle ECD$

$\Rightarrow$ $\angle ACD = 40° + 70° = 110°$

9. $AB \parallel CD$ and EF is a transversal intersecting the parallel lines AB and CD at G and H respectively. Find x and y.



Soln.: $\angle AGH$ and $\angle BGH$ form a linear pair of angles.

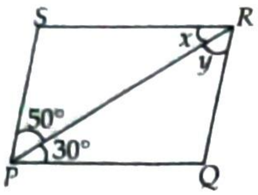
$\therefore$ $x + 120° = 180° \Rightarrow x = 180° - 120° = 60°$

$AB \parallel CD$, EF is a transversal.

$\therefore$ $\angle AGH = \angle GHD$ (Alternate interior angles)

$\Rightarrow$ $x = y = 60°$

10. If $PQ \parallel SR$ and $PS \parallel QR$. Then find the values of x and y.



Soln.: $PQ \parallel SR$ and RP is a transversal.

$\therefore$ $\angle SRP = \angle RPQ$ (Alternate interior angles)

$\Rightarrow$ $x = 30°$

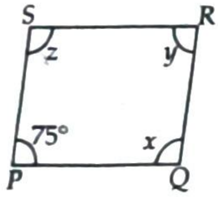
Now, $SP \parallel RQ$ and PR is a transversal.

$\therefore$ $\angle SPR = \angle PRQ$ (Alternate interior angles)

$\Rightarrow$ $50° = y$

Hence, $x = 30°$ and $y = 50°$.

11. Find the values of x, y, z. Given $PQ \parallel SR$ and $PS \parallel QR$.



Soln.: $PQ \parallel SR$ and SP is a transversal. Sum of interior angles on the same side of transversal is 180°.

$\therefore$ $\angle RSP + \angle SPQ = 180° \Rightarrow z + 75° = 180°$

$\Rightarrow$ $z = 180° - 75° = 105°$

Now, $SP \parallel RQ$ and SR is a transversal. Sum of interior angles on the same side of transversal is 180°.

$\therefore$ $\angle PSR + \angle SRQ = 180° \Rightarrow z + y = 180°$

$\Rightarrow$ $y = 180° - 105° = 75°$

Similarly, $SR \parallel PQ$, RQ is a transversal.

$\therefore$ $\angle SRQ + \angle RQP = 180°$ (Co-interior angles)

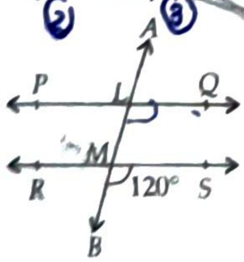
$\Rightarrow$ $y + x = 180° \Rightarrow 75° + x = 180°$

$\Rightarrow$ $x = 180° - 75° = 105°$

$\therefore$ $x = 105°$, $y = 75°$ and $z = 105°$

12. In the given figure, $PQ \parallel RS$ and AB is a transversal intersecting PQ and RS at L and M respectively. If $\angle SMB = 120°$, then find the values of $\angle QLM$, $\angle LMS$, $\angle PLM$.

1 2 3



Soln.: $PQ \parallel RS$ and AB is a transversal.

(i) $\angle QLM = \angle SMB$ (Corresponding angles)

$\Rightarrow$ $\angle QLM = 120°$

(ii) $\angle BMS + \angle LMS = 180°$ (Linear pair)

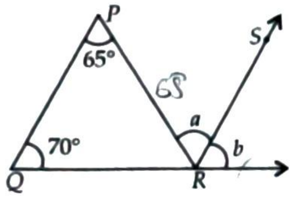
$\Rightarrow$ $120° + \angle LMS = 180°$

$\Rightarrow$ $\angle LMS = 180° - 120° = 60°$

(iii) $\angle PLM = \angle LMS$ (Alternate interior angles)

$\Rightarrow$ $\angle PLM = 60°$

13. In the given figure, $QP \parallel RS$. Find the values of a and b.



Soln.: Since, $QP \parallel RS$ and QR is a transversal.

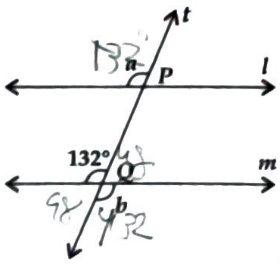
$\therefore$ $b = 70°$ [Corresponding angles]

Now, $QP \parallel RS$ and PR is a transversal.

$\therefore$ $a = 65°$ [Alternate interior angles]

Hence, $a = 65°$ and $b = 70°$.

14. In the given figure, $l \parallel m$ and a line t intersects these lines at P and Q, respectively. Find the sum $2a + b$.



Soln.: Since, $l \parallel m$ and t is a transversal.

$\therefore$ $a = 132°$ [Corresponding angles]

and $b = 132°$ [Vertically opposite angles]

Now, $2a + b = 2 \times 132° + 132°$

$= 264° + 132° = 396°$

15. If the complement of an angle is 62°, then find its supplement.

Soln.: Let the angle be x.

$\therefore$ Its complement $= 90° - x$

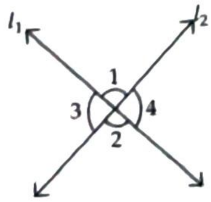
According to question, $90° - x = 62°$

$\Rightarrow$ $90° - 62° = x$ $\Rightarrow$ $x = 28°$

$\therefore$ Supplement of $x = 180° - 28° = 152°$

16. Why are vertically opposite angles always equal? Can you justify this using logic or a diagram?

Soln.: When two lines l_1 and l_2 intersects, four angles are formed.



Now, $\angle 1 + \angle 3 = 180°$ ... (i) (Linear pair)

Also, $\angle 2 + \angle 4 = 180°$ ... (ii) (linear pair)

Similarly,

$\angle 1 + \angle 4 = 180°$ ... (iii)

and $\angle 2 + \angle 3 = 180°$ ... (iv)

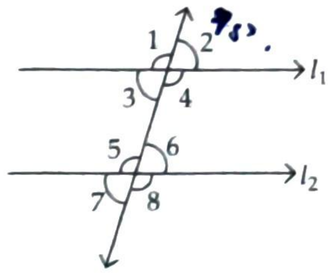
From equations (ii) and (iii), we have

$\angle 2 + \angle 4 = \angle 1 + \angle 4 \Rightarrow \angle 1 = \angle 2$

Similarly, $\angle 3 = \angle 4$

Hence, Vertically opposite angles are always equal.

17. Two lines l_1 and l_2 are cut by a transversal if $\angle 2 = 75°$, then find the values of other angles.



Soln.: We have, $\angle 2 = 75°$

Now, $\angle 2 = \angle 3$ (vertically opposite angles)

$\Rightarrow$ $\angle 3 = 75°$

Now, $\angle 3 = \angle 7$, $\angle 2 = \angle 6$ [Corresponding angles]

$\therefore \angle 2 = \angle 3 = \angle 6 = \angle 7 = 75°$

Now, $\angle 3 + \angle 5 = 180°$ [Co-interior angles]

$\Rightarrow$ $\angle 5 = 180° - 75° = 105°$

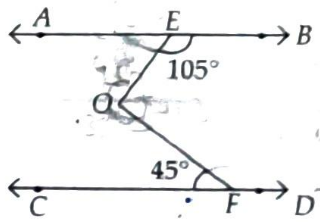
Now, $\angle 5 = \angle 1$ [Corresponding angles]

and $\angle 5 = \angle 8$ [Vertically opposite angles]

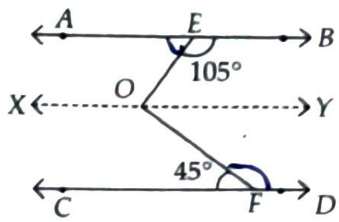
Also, $\angle 8 = \angle 4$ [Corresponding angles]

$\Rightarrow$ $\angle 1 = \angle 5 = \angle 4 = \angle 8 = 105°$

18. In the given figure, $AB \parallel CD$, $\angle BEO = 105°$, $\angle CFO = 45°$. Find the $\angle EOF$.



Soln.: Draw a line XY parallel to AB and CD passing through O, such that $AB \parallel XY$ and $CD \parallel XY$.



$\angle BEO + \angle YOE = 180°$ (Co-interior angles)

$\Rightarrow$ $\angle YOE = 180° - 105° = 75°$

Also, $\angle CFO = \angle YOF$ (Alternate interior angles)

$\therefore$ $\angle YOF = 45°$

Then, $\angle EOF = \angle EOY + \angle FOY = 75° + 45° = 120°$

NCERT Section

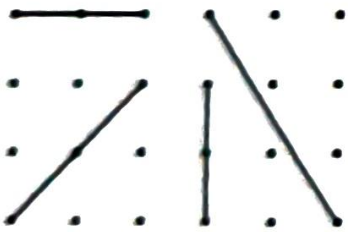
Figure it Out *(Page 108)*

1. List all the linear pairs and vertically opposite angles you observe in figure.

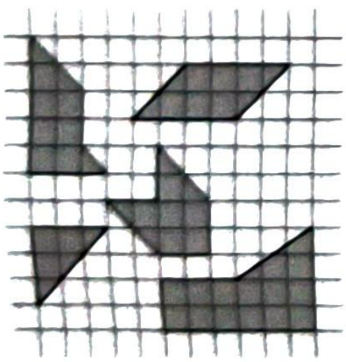
Linear Pairs	$\angle a$ and $\angle b$, ...
Pairs of Vertically Opposite Angles	$\angle b$ and $\angle d$, ...

Figure it Out *(Page 113 & 114)*

1. Draw some lines perpendicular to the lines given on the dot paper in Figure.



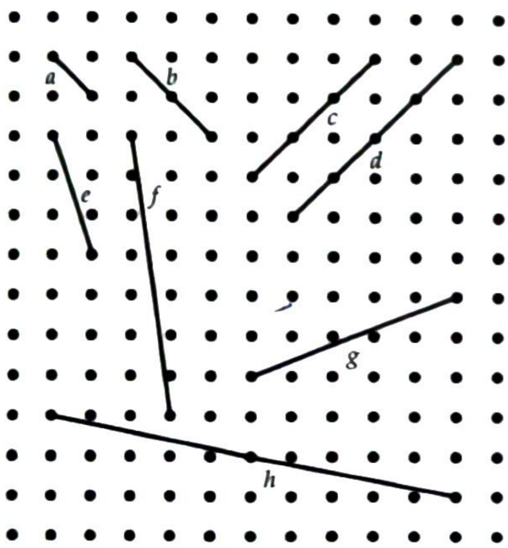
2. In Figure, mark the parallel lines using the notation given above (single arrow, double arrow etc.). Mark the angle between perpendicular lines with a square symbol.



(a) How did you spot the perpendicular lines?
(b) How did you spot the parallel lines?

3. In the dot paper, draw different sets of parallel lines. The line segments can be of different lengths but should have dots as endpoints.

4. Using your sense of how parallel lines look, try to draw lines parallel to the line segments on this dot paper.



(a) Did you find it challenging to draw some of them?
(b) Which ones?
(c) How did you do it?

5. In Figure, which line is parallel to line a — line b or line c? How do you decide this?

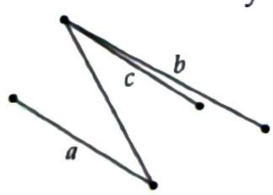


Figure it Out *(Page 119)*

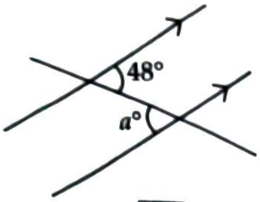
1. Can you draw a line parallel to l, that goes through point A? How will you do it with the tools from your geometry box? Describe your method.

Figure it Out

(Page 123-125)

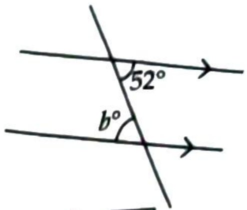
1. Find the angles marked below.

(i)



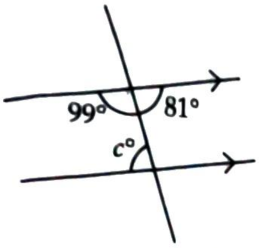
a = ☐

(ii)



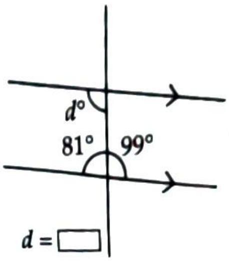
b = ☐

(iii)



c = ☐

(iv)

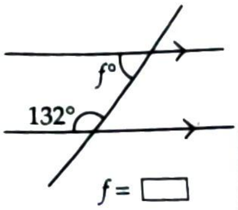


d = ☐

(v)

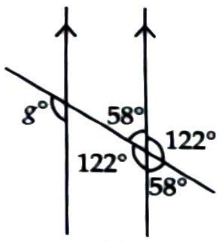
e = ☐

(vi)



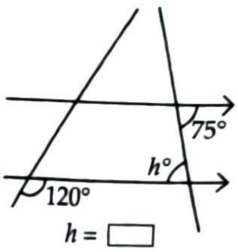
f = ☐

(vii)



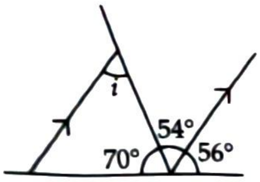
g = ☐

(viii)



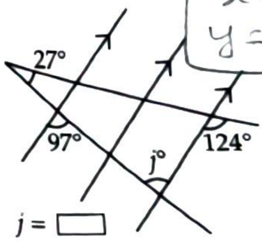
h = ☐

(ix)



i = ☐

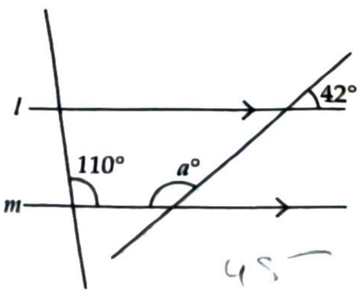
(x)



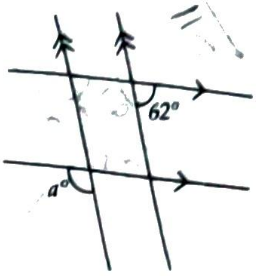
j = ☐

2. Find the angle represented by *a*.

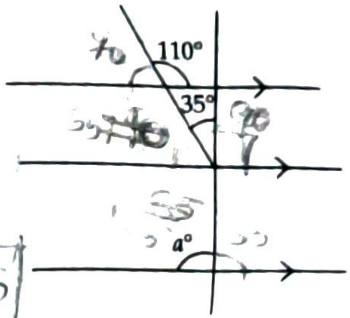
(i)



(ii)

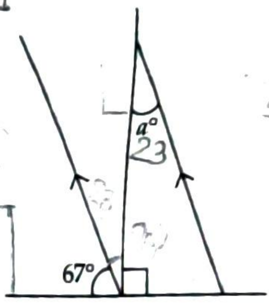


(iii)



a=125

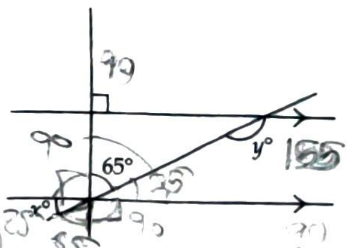
(iv)



a=23

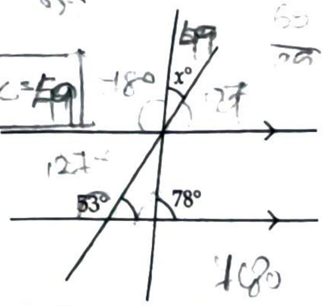
3. In the figures below, what angles do *x* and *y* stand for?

(i)



x=25°
y=155

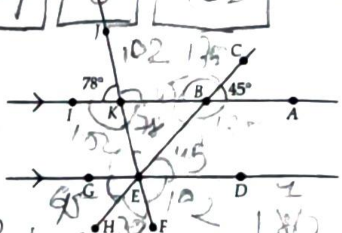
(ii)



x=49

4. In Figure, ∠ABC = 45° and ∠IKJ = 78°. Find angles ∠GEH, ∠HEF, ∠FED.

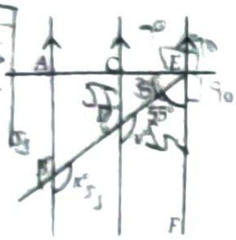
47° | 32° | 102°



5. In Figure, AB is parallel to CD and CD is parallel to EF. Also, EA is perpendicular to AB. If $\angle BEF = 55°$, find the values of x and y.

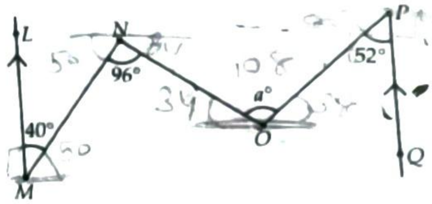
x = 55

y = 55



6. What is the measure of angle $\angle NOP$ in Figure?

a = 108°



[Hint : Draw lines parallel to LM and PQ through points N and O.]

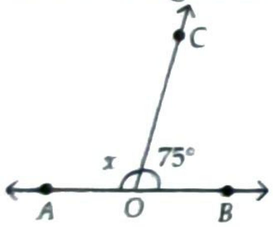
Exercise

Multiple Choice Questions

LEVEL - 1

Intersecting Lines

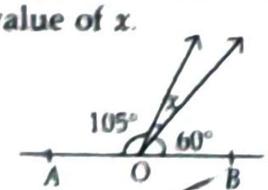
1. In the adjoining figure, the value of x such that $\overline{AOB}$ is a line segment is



(a) 40°
(b) 55°
(c) 105°
(d) 180° *(NCERT Page 107)* U

2. The supplement of an acute angle is ______ angle.
(a) Acute
(b) Obtuse
(c) Right
(d) Straight *(Page 107)* R

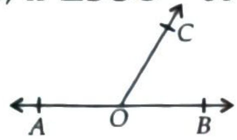
3. In the figure, if AOB is a straight line, then find the value of x.



(a) 40° (b) 15°
(c) 20° (d) 50° *(Page 107)* U

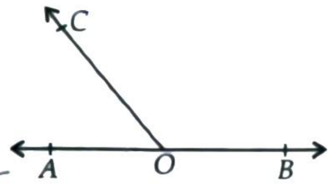
4. If an angle is its own supplementary angle, then its measure is
(a) 30° (b) 45° (c) 60° (d) 90°

5. Find $\angle AOC$, if $\angle BOC = 60°$.



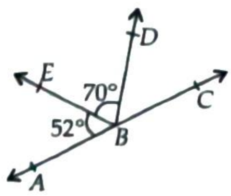
(a) 120° (b) 90°
(c) 60° (d) 45° *(Page 107)* U

6. In the given figure, AOB is straight line and the ray OC stands on it. If $\angle BOC = 130°$, then find $\angle AOC$.



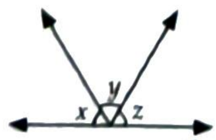
(a) 50° (b) 70°
(c) 20° (d) 160° *(Page 107)* U

7. In the figure, ABC is a straight line. Find $\angle CBD$.



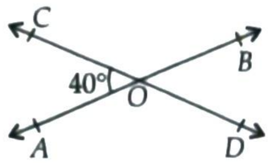
(a) 58° (b) 62°
(c) 73° (d) 25° *(Page 107)*

8. In the given figure, $x = y = z$. Find $y + z$.



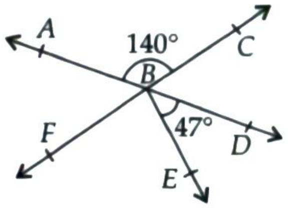
(a) 60° (b) 120°
(c) 130° (d) 100° *(Page 107)* U

9. In the given figure, two straight lines AB and CD intersect at a point O. If $\angle AOC = 40°$, then $\angle BOD$ = ?



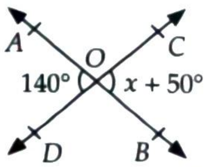
(a) 140° (b) 50°
(c) 40° (d) 160° *(Page 108)* U

10. In the figure, AD and CF are straight lines. Find $\angle EBF$.



(a) 93° (b) 140°
(c) 47° (d) 107° *(Page 108)* U

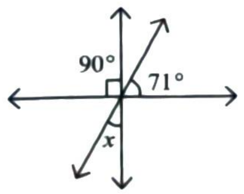
11. In the given figure, two lines AB and CD intersect each other at O. Find the value of x.



(a) 120° (b) 90°
(c) 50° (d) 70° *(Page 108)*

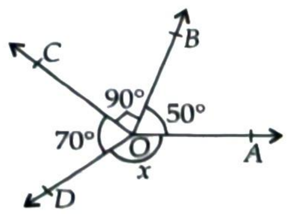
12. What is the value of x in the given figure?

(a) 71°
(b) 90°
(c) 19°
(d) 24°



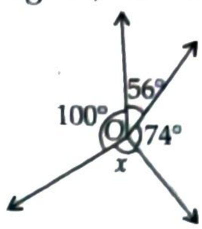
(Page 109) U

13. In the given figure, rays OA, OB, OC and OD are such that $\angle AOB = 50°$, $\angle BOC = 90°$, $\angle COD = 70°$ and $\angle AOD = x$. The value of x is



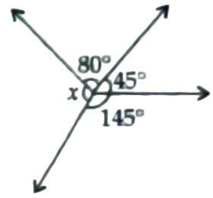
(a) 50° (b) 70°
(c) 150° (d) 90° *(Page 109)*

14. In the given figure, the value of x is



(a) 130°
(b) 56°
(c) 100°
(d) 74° U

15. Find the value of x in the figure given below.



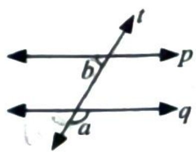
(a) 90° (b) 45°
(c) 80° (d) 145° U

16. Two angles are complementary and equal, then find each of the angles.
(a) 90°, 90° (b) 45°, 45°
(c) 180°, 180° (d) 135°, 45° Ap

17. The complement of an angle of 80° measures
(a) 100° (b) 10° (c) 280° (d) 20°

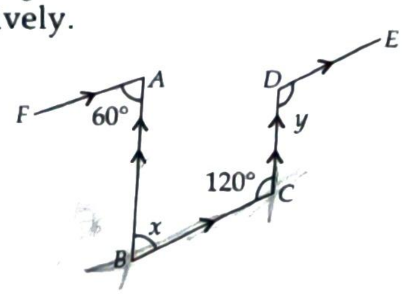
Parallel Lines

18. In the given figure, $p \parallel q$ and t is transversal. Find the value of a and b (respectively), if $a = 3b$.



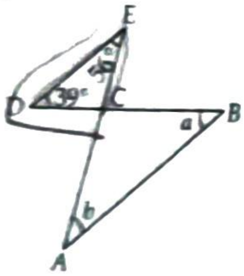
(a) 135°, 45°
(b) 130°, 90°
(c) 155°, 55°
(d) 45°, 135° *(Page 121)* U

19. In the figure, find the value of x and y respectively.



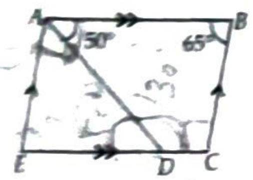
(a) 60°, 120°
(b) 70°, 110°
(c) 50°, 130°
(d) 80°, 100° (Page 122)

20. In the figure, $DE \parallel AB$, $\angle DEC = 56°$ and $\angle EDC = 39°$. Find the values of a and b respectively.



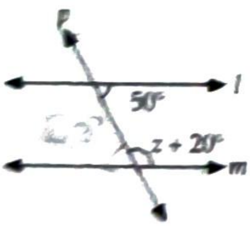
(a) 56°, 39°
(b) 85°, 56°
(c) ~~39°, 56°~~
(d) 39°, 85° (Page 122)

21. In the figure, *ABCE* is a parallelogram. Find $\angle ADC$.



(a) 130°
(b) 150°
(c) 50°
(d) 115° (Page 122)

22. In the given figure, transversal *p* cuts parallel lines *l* and *m*. Find the value of z.



(a) 110°
(b) 70°
(c) 220°
(d) 120° (Page 123) U

23. Statements *a* and *b* are as given below:

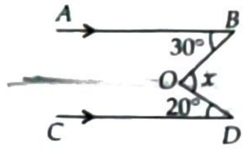
a : If two lines intersect, then the vertically opposite angles are equal.

b : If a transversal intersects two other lines, then the sum of two interior angles on the same side of the transversal is 180°.

Then

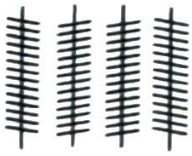
(a) Both *a* and *b* are true
(b) *a* is true and *b* is false
(c) *a* is false and *b* is true
(d) both *a* and *b* are false (Page 123) Ap

24. In the following figure, find the value of *x*.



(a) 30°
(b) 20°
(c) 10°
(d) 50° (Page 125)

25. Which of the given lines are parallel?

(a) 

(b) 

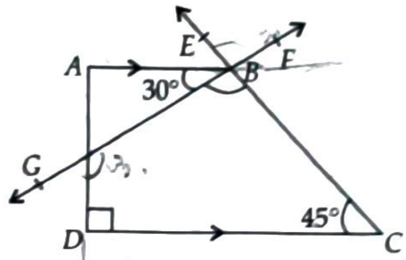
(c) 

(d) ~~All of these~~

(Page 125)

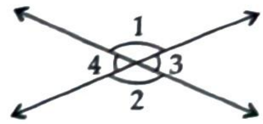
LEVEL - 2

26. In the figure, *ABCD* is a trapezium. *FG* is a straight line. Find $\angle EBF$.



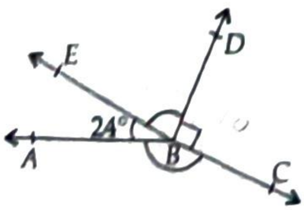
(a) 70°
(b) 105°
(c) 150°
(d) 60°

27. Which of the following is false?



(a) $\angle 1 = \angle 2$
(b) $\angle 1 + \angle 3 = 180°$
(c) $\angle 1 = \angle 4$
(d) $\angle 3 = \angle 4$ U

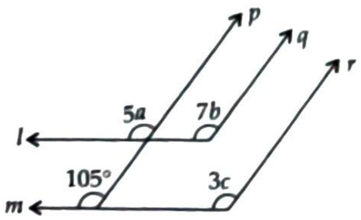
28. In the figure, $\angle CBD$ is a right angle. $\angle DBE$ is thrice of $\angle ABE$. Find $\angle ABC$.



(a) 142°
(b) 124°
(c) 147°
(d) 174° U

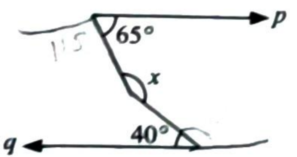
29. If the angle formed by hour hand and a third pointer is 65°, and hour and minute hands form a straight line, what is the angle between the minute hand and the third pointer?
(a) 105° (b) 115° (c) 125° (d) 135°

30. In the given figure, $p \parallel q \parallel r$ and $l \parallel m$. Find the value of $b + c$.



(a) 50° (b) 120°
(c) 70° (d) 30°

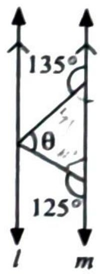
31. In the given figure $p \parallel q$. What is the measure of x?



(a) 165° (b) 105° (c) 155° (d) 175°

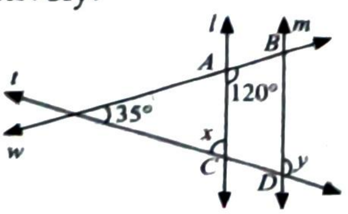
32. In the given figure $l \parallel m$. Find the measure of θ.

Theta



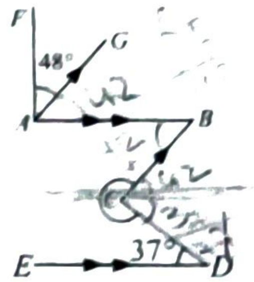
(a) 100° (b) 80°
(c) 70° (d) 160° U

33. In the following figure, lines l and m are parallel to each other. t and w are two transversals intersecting the lines at A and B, and C and D. Find the values of x and y respectively.



(a) 95°, 85° (b) 60°, 120°
(c) 85°, 95° (d) 75°, 85°

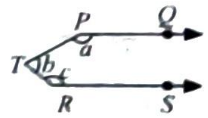
34. Given, $AB \parallel ED$, $AG \parallel CB$ and $AF \perp AB$. $\angle FAG = 48°$, $\angle CDE = 37°$. Find the value of x.



(a) 263° (b) 277°
(c) 281° (d) 308° U

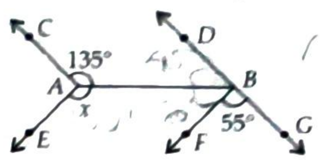
35. Draw four rays OA, OB, OC, OD such that $\angle AOB = 58°$, $\angle COD = 64°$. OP bisects $\angle AOB$, OQ bisects $\angle COD$. If OQ is perpendicular to OP, then find $\angle BOC$.
(a) 29° (b) 50°
(c) 51° (d) 48° Ev

36. If ray PQ and RS are parallel as shown in the figure, then find $a + b + c$.



(a) 180° (b) 120° (c) 360° (d) 185°

37. In the given figure $AC \parallel BD$ and $AE \parallel BF$. The measure of x is



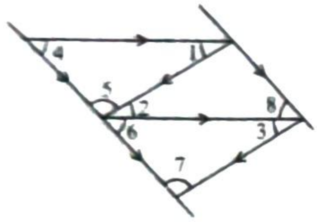
(a) 130° (b) 100°
(c) 70° (d) 50° U

38. In a paper folding activity, a student folds a sheet such that two creases are perpendicular to the same third crease. What can be said about the two creases?
(a) They are intersecting but not necessarily parallel.
(b) They must be perpendicular to each other.
(c) They are always parallel.
(d) They form supplementary angles.

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

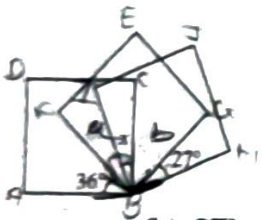
39. Which of the following options is incorrect?



(a) $\angle 1 = \angle 3$
(b) $\angle 8 = \angle 6$
(c) $\angle 1 + \angle 3 = 180°$
(d) None of these

U

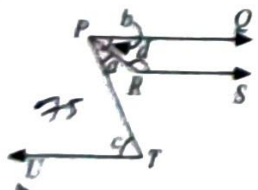
40. The given figure shows three identical squares. Find x.



(a) 30°   (b) 27°
(c) 36°   (d) 16°

U

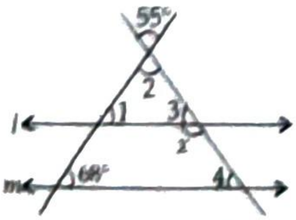
41. In the given figure, PQ, RS and UT are parallel lines. If $c = 75°$ and $a = \frac{2}{5}c$, then find $b + \frac{d}{2}$.



(a) 92°   (b) 115°
(c) 112.5°   (d) 135.5°

U

42. Given $l \parallel m$, find the value of x in the following figure.



(a) 96°   (b) 89°
(c) 123°   (d) 121°

43. Find the value of $x + y + z$ in Fig. (i) and Fig. (ii) respectively.

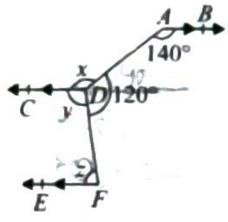


Fig. (i)

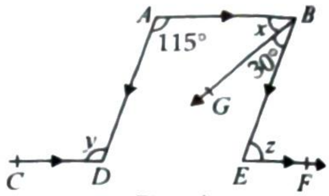
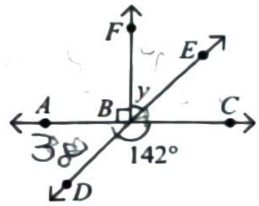


Fig. (ii)

(a) 225°, 320°   (b) 320°, 225°
(c) 320°, 215°   (d) 215°, 320°

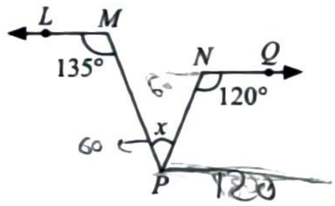
U

44. In the given figure (not drawn to scale), ABC and DBE are straight lines. Find the value of y.



(a) 38°   (b) 142°   (c) 52°   (d) 68°

45. Given that $LM \parallel NQ$, $\angle LMP = 135°$ and $\angle QNP = 120°$. Find the value of x.



(a) 60°   (b) 75°
(c) 135°   (d) 120°

U

46. Fill in the blanks and select the correct option.

(i) A line cutting two parallel lines making pairs of corresponding and alternate angles is called __P__.

(ii) If $\angle m = 82°$ and $\angle n = 98°$, then $\angle m$ and $\angle n$ are __Q__ angles.

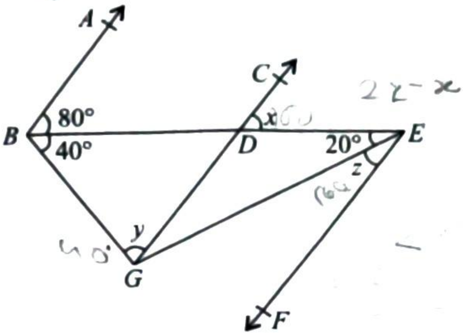
(iii) An angle which is half of its supplement is of measure __R__.

(iv) The angle $(x - 50°)$ and $(140° - x)$ are __S__ angles.

	P	Q	R	S
(a)	Parallel	Supplementary	30°	Complementary
(b)	Transversal	Complementary	60°	Supplementary
(c)	Parallel	Complementary	30°	Supplementary
(d)	Transversal	Supplementary	60°	Complementary

Ap

47. In the given figure (not drawn to scale), if $AB \parallel CG \parallel EF$, then find
(i) $2z - x$
(ii) $x + 2y$



	(i)	(ii)		(i)	(ii)
(a)	40°	220°	(b)	60°	200°
(c)	50°	200°	(d)	40°	200°

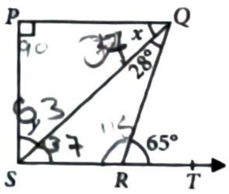
U

48. Fill in the blanks and select the correct option.
(i) If two lines intersect each other, then the vertically opposite angles are __P__.
(ii) The angles of a linear pair are __Q__ as well as __R__.
(iii) If a transversal is perpendicular to two parallel lines then sum of each pair of corresponding angles is __S__.

	P	Q	R	S
(a)	Unequal	adjacent	complementary	90°
(b)	Unequal	complementary	supplementary	90°
(c)	Equal	adjacent	supplementary	180°
(d)	Equal	supplementary	complementary	180°

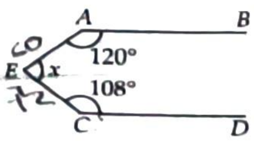
Ap

49. In figure, $PQ \perp PS$, $PQ \parallel SR$, $\angle SQR = 28°$ and $\angle QRT = 65°$, then x and y will be



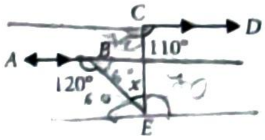
(a) $x = 37°, y = 53°$ (b) $x = 53°, y = 37°$
(c) $x = 57°, y = 33°$ (d) $x = 33°, y = 57°$

50. In the given figure $AB \parallel CD$. Then the value of x is



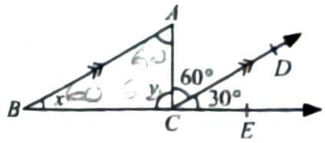
(a) 220° (b) 140°
(c) 150° (d) 132°

51. In given figure, $AB \parallel CD$, $\angle ABE = 120°$, $\angle DCE = 110°$ and $\angle BEC = x$, then x will be



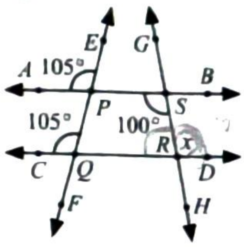
(a) 60° (b) 50°
(c) 40° (d) 70°

52. In the given figure, $\angle DCA = 60°$, $\angle ECD = 30°$, then find x and y.



(a) 60°, 30° (b) 30°, 90°
(c) 30°, 60° (d) 60°, 60°

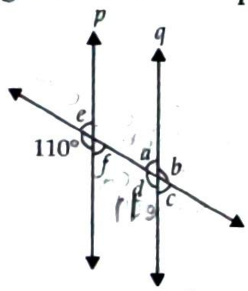
53. In the given figure, find the value of x.



(a) 80° (b) 75°
(c) 100° (d) 105°

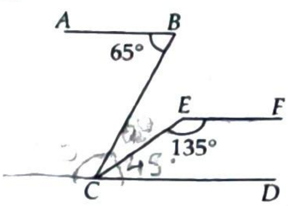
U

54. In the adjoining figure, $p \parallel q$. Find the unknown angles c and d respectively.



(a) 70°, 110° (b) 110°, 70°
(c) 80°, 100° (d) 100°, 80°

55. In the given figure, $AB \parallel CD$ and $CD \parallel EF$, find $\angle BCE$.



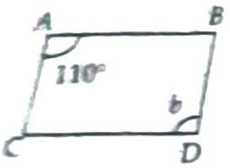
(a) 25° (b) 30°
(c) 20° (d) 40°

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which, one is correct.

1. Match the following :

List-I	List-II
(P) If one angle of a linear pair is obtuse then measure of other one can be	1. 110°
(Q) Diagonals of a square intersect at	2. 90°
(R) If $AB \parallel CD$ and $AC \parallel BD$, then value of b is	3. 45°
(S) Measure of two vertically opposite angles are $x + 10°$ and $110°$, then value of x is	4. 100°



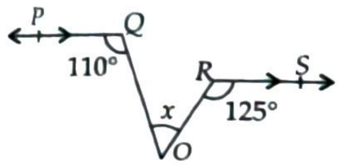
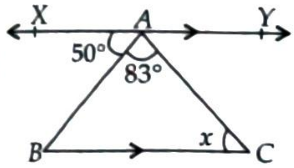
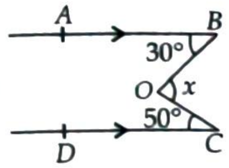
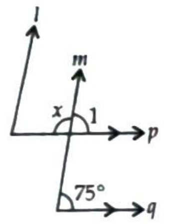
(a) P-1, Q-2, R-3, S-4 (b) P-3, Q-2, R-1, S-4
(c) P-1, Q-4, R-3, S-2 (d) P-3, Q-4, R-2, S-1

2. Match the following :

List-I	List-II
(P) Two angles whose sum is 90° are called	1. Adjacent angles
(Q) Two angles whose sum is 180° are called	2. Complementary angles
(R) Two angles that are formed by two intersecting lines, which are not adjacent are called	3. Supplementary angles
(S) Two angles with a common vertex, a common arm and the other arms lying on the opposite sides of the common arm form a pair of	4. Vertically opposite angles

(a) P-1, Q-2, R-3, S-4 (b) P-3, Q-1, R-4, S-2
(c) P-2, Q-3, R-4, S-1 (d) P-4, Q-2, R-1, S-3

3. Match the value of x given in List-II with the figure given in List-I.

List-I	List-II
(P) 	1. $x = 47°$
(Q) 	2. $x = 55°$
(R) 	3. $x = 105°$
(S) 	4. $x = 80°$

(a) P-1, Q-2, R-3, S-4 (b) P-3, Q-1, R-4, S-2
(c) P-2, Q-1, R-4, S-3 (d) P-4, Q-2, R-1, S-3

Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

(a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If assertion is false but reason is true.

1. **Assertion :** If two lines intersect, then the vertically opposite angles are equal.
   Reason : Sum of all the angles around a point is 180°.

2. **Assertion** : The complement of 35° is 55°.
Reason : If two angles are complementary, then their sum is 90°.

3. **Assertion** : The supplement of $\frac{1}{2}$ of 120° is 60°.
Reason : If two angles are supplementary then their sum is 180°.

4. **Assertion** : Two parallel lines do not intersect each other.
Reason : The distance between two parallel lines is the same everywhere, *i.e.*, it neither increases nor decreases.

5. **Assertion** : A transversal intersects two parallel lines then all 8 angles formed can be equal.
Reason : Transversal intersects the parallel lines at right angles.

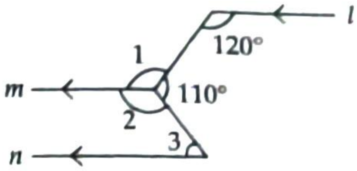
6. **Assertion** : A pair of lines can be perpendicular without being intersecting.
Reason : Perpendicularity means two lines have an angle of 90°.between them.

7. **Assertion** : While drawing parallel lines with a set square, the distance between the lines can be fixed without measuring it explicitly.
Reason : The distance between the lines depends only on how far the set square is slide along the ruler.

8. **Assertion** : Optical illusions can trick the eye but not the properties of geometry.
Reason : In geometry, definition of parallel lines are independent of visual appearance.

Comprehension Type

PASSAGE-I : In the given figure l, m and n are parallel.

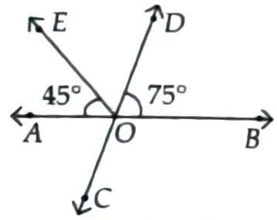


1. Find the measure of ∠1.
(a) 60° (b) 120°
(c) 125° (d) none of these

2. What will be the measure of ∠2?
(a) 110° (b) 120°
(c) 130° (d) 140°

3. Find the measure of ∠3.
(a) 50° (b) 60°
(c) 70° (d) 80°

PASSAGE-II : In the given figure, AB is a straight line. $\angle BOD = 75°$ and $\angle AOE = 45°$.



1. Find the measure of $\angle EOD$.
(a) 45° (b) 50°
(c) 55° (d) 60°

2. What will be the measure of $\angle COB$?
(a) 105° (b) 110°
(c) 125° (d) 130°

3. Find the value of $\angle COB - \angle AOC$.
(a) 0° (b) 30°
(c) 60° (d) 90°

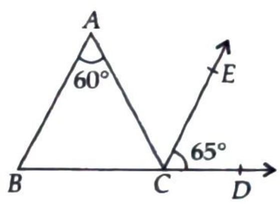
Subjective Problems

Very Short Answer Type

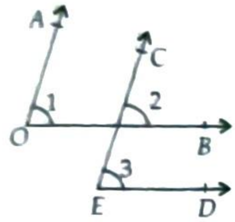
1. In an optical illusion, how can you confirm whether two lines are truly parallel?

2. How do you shift the set square to ensure the new line remains parallel to the original?

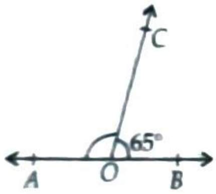
3. In the adjoining figure, it is given that $\angle A = 60°$, $CE \parallel BA$ and $\angle ECD = 65°$. Find $\angle ACB$.



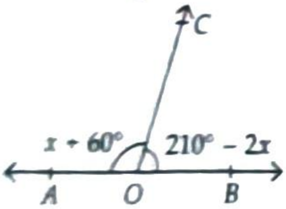
4. In the adjoining, figure, it is given that $OA \parallel EC$ and $OB \parallel ED$. Prove that $\angle AOB = \angle CED$.



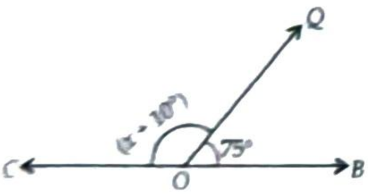
5. In the given figure, AB is a straight line and $\angle BOC = 65°$, find the measure of $\angle AOC$.



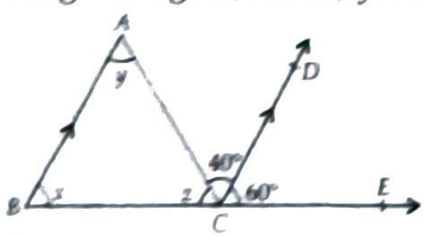
6. In the adjoining figure, what value of x will make AOB a straight line?



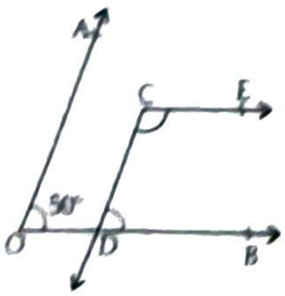
7. In the figure, COB is a straight line. Find the value of x.



8. In the given figure, find x, y and z.



9. In the adjoining figure, it is being given that $AO \parallel CD$, $OB \parallel CE$ and $\angle AOB = 50°$.



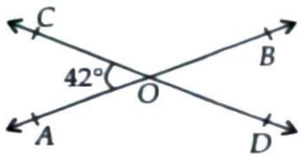
Find the measure of $\angle ECD$.

10. Prove that the two lines in a plane, which are perpendicular to the same given line in the plane, are parallel to each other.

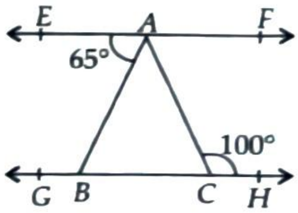
Short Answer Type

1. Prove that two lines which are parallel to the same given line are parallel to each other.

2. In the given figure, two straight lines AB and CD intersect at a point O.
If $\angle AOC = 42°$, then find the measure of each of the angles given below.
(i) $\angle AOD$ (ii) $\angle BOD$
(iii) $\angle COB$

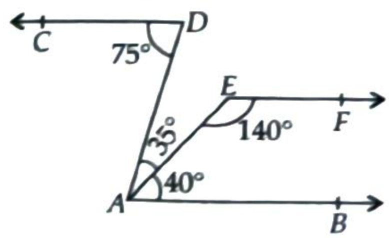


3. In the given figure, $EF \parallel GH$, $\angle EAB = 65°$ and $\angle ACH = 100°$.

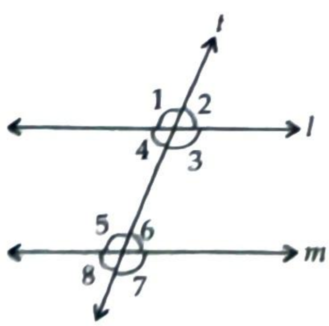


Determine : (i) $\angle BAC$ (ii) $\angle CAF$.

4. In the given figure, show that $CD \parallel EF$.



5. In the given figure, $l \parallel m$ and t is a transversal such that $\angle 1 = 135°$.

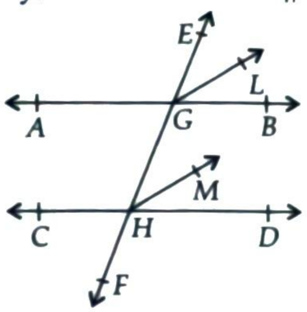


Find the measure of each of the angles $\angle 2$, $\angle 3$, $\angle 4$, $\angle 5$, $\angle 6$, $\angle 7$ and $\angle 8$.

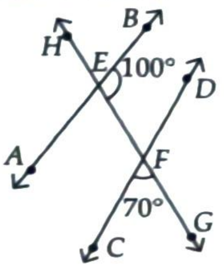
6. Find the complement of each of the following angles :
(i) 60° (ii) 25° (iii) 72°

7. Find the supplement of each of the following angles :
(i) 125° (ii) 64° (iii) 38°

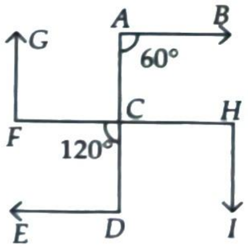
8. In the adjoining figure, $AB \parallel CD$ and EF is a transversal intersecting them at G and H respectively. If GL and HM are the bisectors of the corresponding angles EGB and EHD respectively, show that $GL \parallel HM$.



9. In the given figure, $\angle BEF = 100°$ and $\angle CFG = 70°$. Check whether line AB and CD are parallel or not.

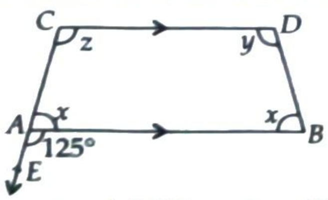


10. In the given figure (not drawn to scale), if $AB \parallel FH \parallel ED$ and $GF \parallel AD \parallel HI$ then find the measure of $\angle CHI$ and $\angle ACH$.



Long Answer Type

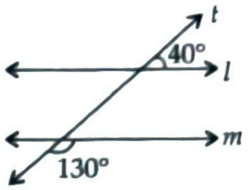
1. In the given figure, $AB \parallel CD$ and CA has been produced to E so that $\angle BAE = 125°$.



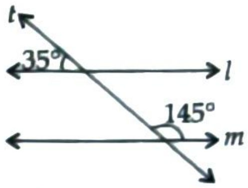
If $\angle BAC = x$, $\angle ABD = x$, $\angle BDC = y$ and $\angle ACD = z$, then find the value of $y + z - x$.

2. In each of the given figure, two lines l and m are cut by a transversal t. Check whether $l \parallel m$ or not.

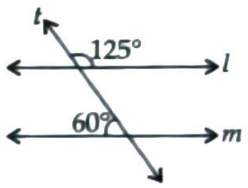
(i)



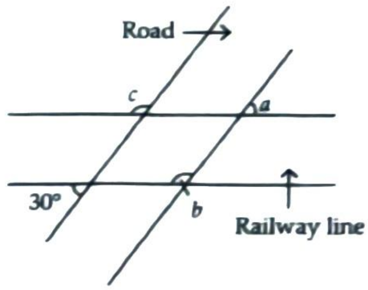
(ii)



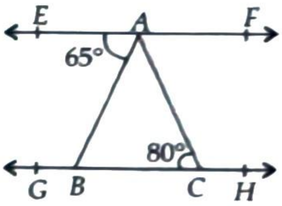
(iii)



3. A road crosses a railway line at an angle of 30° as shown in figure. Find the values of a, b and c.

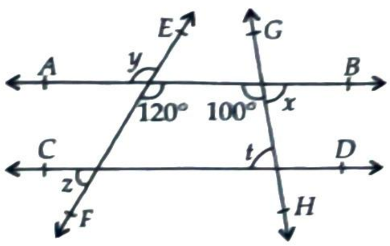


4. In the given figure, $EF \parallel GH$.



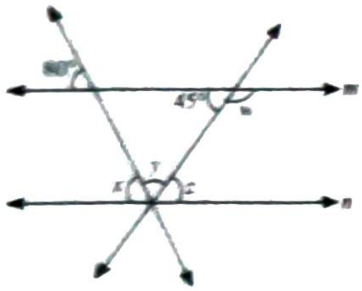
Determine :
(i) $\angle ABC$ (ii) $\angle ACH$
(iii) $\angle BAC$ (iv) $\angle CAF$.

5. In the following figure, $AB \parallel CD$. Find the unknown angles.

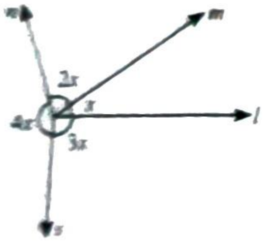


Integer/Numerical Value Type

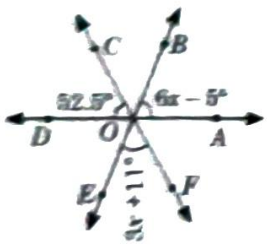
1. If $m \parallel n$, then the measure (in degrees) of $z - y$ is ________.



2. In the given figure, the sum of acute angles (in degrees) is ________.

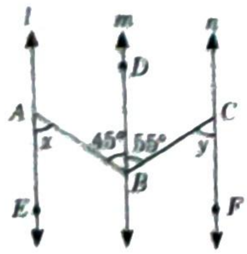


3. In the given figure, three lines intersect at O. Find the measure of $\angle BOC$ (in degrees).



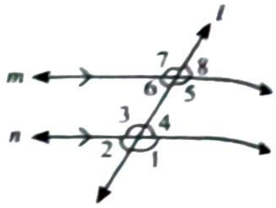
4. The difference of two supplementary angles (in degrees) A and B such that $A = (3x - 25)°$ and $B = (2x + 5)°$ is ________.

5. In the given figure, $l \parallel m \parallel n$, $\angle ABD = 45°$ and $\angle CBD = 55°$. Find the value of $y - x$.



Case Based Questions

Case I : Riya and Rohit are sitting on a wooden plank over a canal in their field. They observe that the wooden plank acts as a transversal, with the canal's banks acting as parallel lines.



Based on the above information, answer the following questions. CFQs

1. Which of the following pairs represent alternate exterior angles?
(a) $\angle 6$ and $\angle 4$ (b) $\angle 8$ and $\angle 2$
(c) $\angle 8$ and $\angle 1$ (d) $\angle 7$ and $\angle 4$

2. Which of the following angles is same as that of $\angle 5$?
(a) $\angle 3$ (b) $\angle 1$
(c) $\angle 7$ (d) All of these

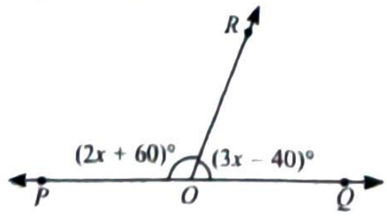
3. If $\angle 2 = 75°$, then the measure of $\angle 6 + \angle 1$ is
(a) 150° (b) 180°
(c) 210° (d) none of these

4. Which of the following angles are not supplementary?
(a) $\angle 4$ and $\angle 5$ (b) $\angle 1$ and $\angle 7$
(c) $\angle 3$ and $\angle 6$ (d) $\angle 7$ and $\angle 8$

5. If $\angle 1 + \angle 4 = 3x$, then measure of $2x$ is
(a) 60° (b) 120°
(c) 75° (d) 150°

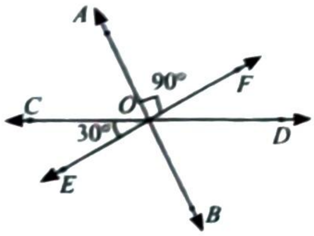
Case II : Sheetal was helping her brother Anupam in understanding the concept of special types of angles. She used some match sticks to make various figures so that Anupam understood the application of various angles. Answer the following questions which are framed by Sheetal for Anupam. CFQs

1. In the adjoining figure, the value of x is



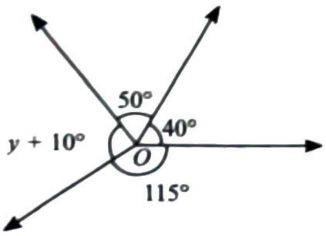
(a) 140° (b) 15°
(c) 32° (d) 20°

2. In the adjoining figure, AB, CD and EF are straight lines. If $\angle AOF = 90°$ and $\angle COE = 30°$, then the measure of $\angle AOC$ is



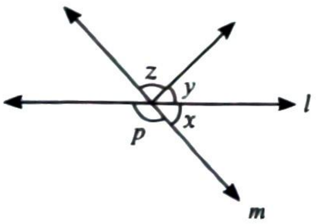
(a) 105° (b) 120° (c) 60° (d) 115°

3. In the given figure, the value of y is



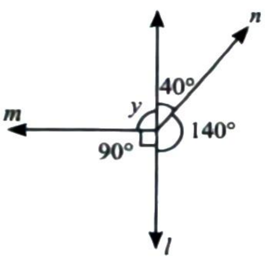
(a) 155° (b) 130° (c) 160° (d) 145°

4. In the given figure, if $x = 15°$ and $z = 105°$, then the measure of angle p is



(a) 165° (b) 105°
(c) 75° (d) 120°

5. The complement of y in the given figure is



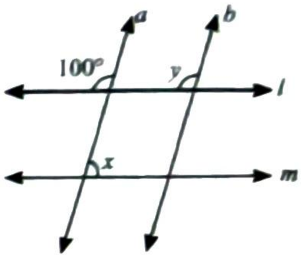
(a) 90° (b) 180° (c) 60° (d) 0°

Case III : Manya's maths teacher had taught the concept of "Lines and Angles". She wrote the list of angles formed when a transversal intersected a pair of parallel lines, in order to take a surprise test.

- Corresponding angles are equal
- Alternate interior angles are equal
- Co-interior angles are supplementary
- Alternate exterior angles are equal

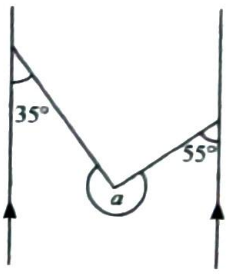
Now, answer the following questions asked by Manya's math teacher. **CFQs**

1. In the adjoining figure, if $a \parallel b$ and $l \parallel m$, then the value of $x + y$ is



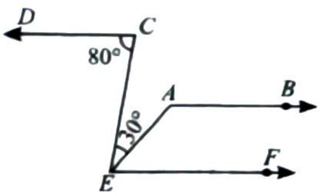
(a) 200° (b) 160°
(c) 180°
(d) Can't be determined

2. In the given figure, the measure of angle a is



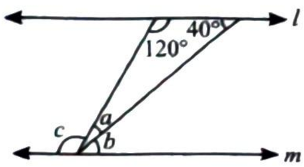
(a) 110° (b) 90° (c) 270° (d) 215°

3. In the given figure, if $AB \parallel CD \parallel EF$, then the measure of $\angle EAB$ is



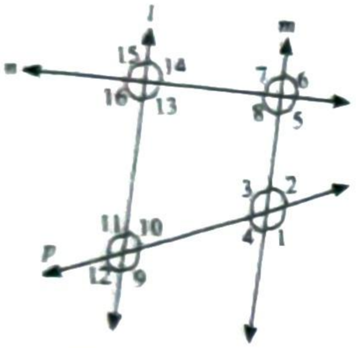
(a) 50° (b) 130° (c) 100° (d) 80°

4. Which of the following is correct about the given figure?



(a) $a + c = 5b$ (b) $c = 2a + 3b$
(c) $b = c + a$ (d) $2(a + b) = c$

5. In the given figure, if $l \parallel m$, then which of the following is incorrect?



(a) $\angle 15 = \angle 5$ (b) $\angle 10 = \angle 2$
(c) $\angle 9 = \angle 3$ (d) $\angle 10 = \angle 14$

Case IV : While preparing for Olympiad Exam, Amit come across the topic complementary and supplementary angles with which he always had a problem. Help him in answering the following questions which he is not able to solve by himself. CFQs

1. Find the angle which is two-third of its supplement.
(a) 72° (b) 108° (c) 84° (d) 96°

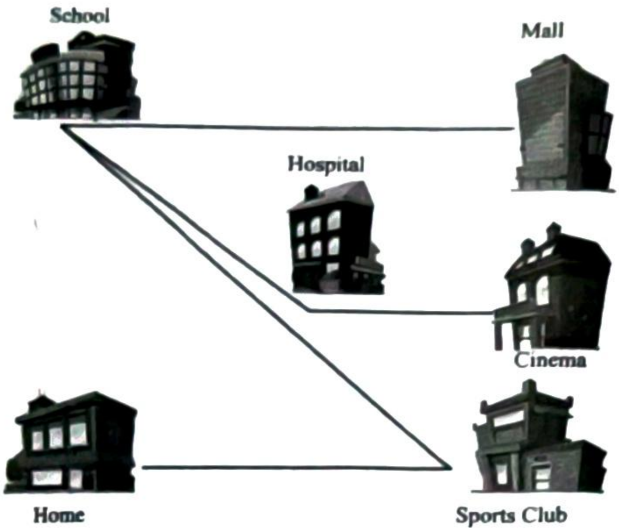
2. If complement of an angle is $\frac{3}{8}$ of its supplement, then find the angle.
(a) 42° (b) 36° (c) 48° (d) 54°

3. If an angle is equal to its complementary angle, then the measure of angle is
(a) 30° (b) 45° (c) 60° (d) 50°

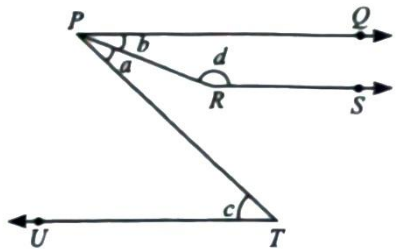
4. If the sum of two supplementary angles is $6x$, then measure of x is
(a) 20° (b) 15° (c) 30° (d) 60°

5. If complement of an angle is 9° less than twice the angle, then measure of the angle is
(a) 57° (b) 54° (c) 33° (d) 36°

Case V : A surveyor represented the roads connecting various land marks near by his home in the locality as shown below.



His son observed the map and draw parallel lines with transversal.



Based on the above data, answer the following questions. CFQs

1. If $\angle b$ is 5/4 times $\angle c$ and they are supplements of each other, then $\angle c$ is
(a) 50° (b) 130° (c) 80° (d) 65°

2. If $\angle c = 80°$, then $\angle a + \angle b =$
(a) 21° (b) 131° (c) 80° (d) 100°

3. If $\angle a$ is one-fourth of complement of $\angle d$, then the measure of $\angle b$ is
(a) $3\angle a$ (b) $4\angle a + 90°$
(c) $\angle a + 50°$ (d) $90° + \angle a$

4. If $\angle c = 60°$, then which of the following is correct?
(a) $\angle d + 2\angle a = 210°$ (b) $3\angle d - \angle a = 150°$
(c) $\angle d - \angle a = 120°$ (d) $\angle d - \angle c = 60°$

5. If $\angle d = 142°$, $\angle a = \frac{\angle c}{3}$, then the measure of $\angle c$ is
(a) 42° (b) 36° (c) 54° (d) 57°